

220108037

Assumptions

Given to us is that $V_{th,n} = 0.37V$ $V_{dd} = 1.8V$

$$V_{th,p} = 0.39V$$

$$L_{min} = 0.18\mu m$$

$$\mu_n C_{ox} = 230\mu A/V^2$$

$$W_{min} = 0.27\mu m$$

$$\mu_p C_{ox} = 100\mu A/V^2$$

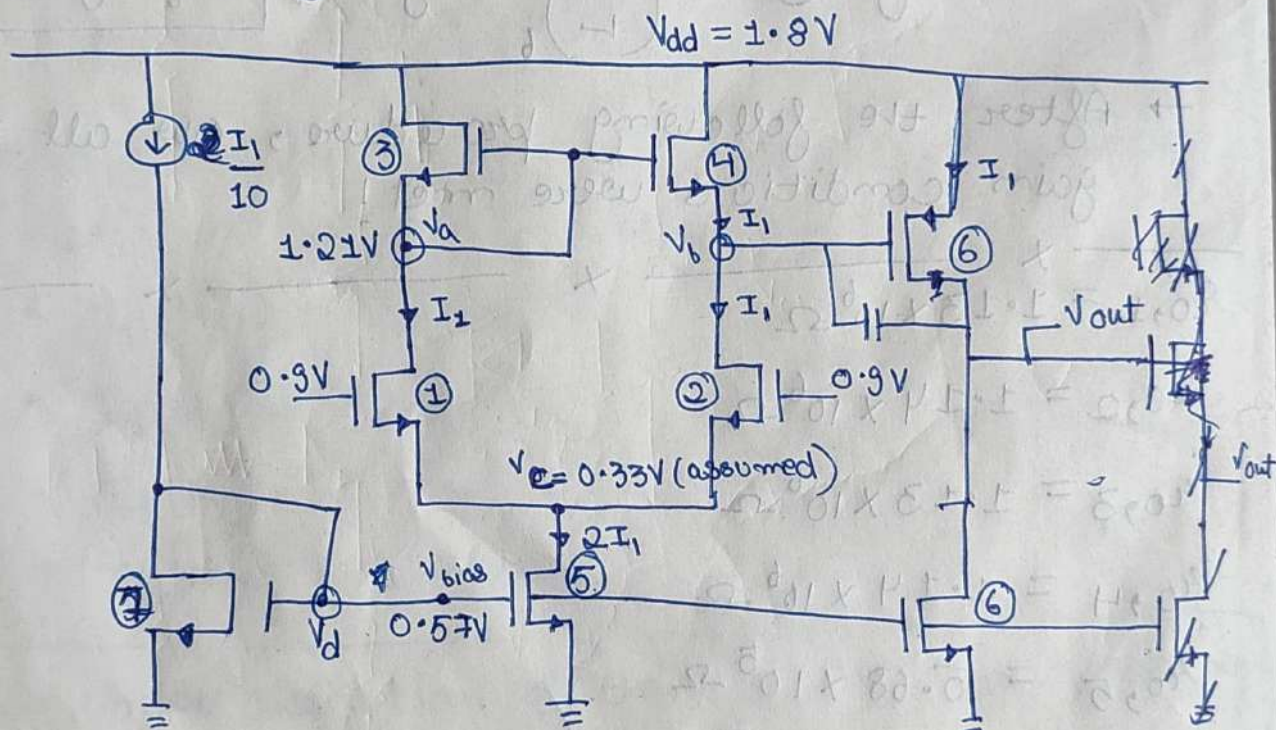
→ Given to us is DC gain = 40db

$$\Rightarrow \boxed{\text{Open Loop gain} = 100}$$

→ We have to design an Op-Amp with gain 2.

→ Gate Overdrive = $0.2V$, $\lambda_n = \lambda_p = 0.1$

⇒ Circuit Diagram



→ For max swing we take input to be $V_{DD}/2$.

We are neglecting ~~Channel length Modulation~~ and Body effect i.e. $\lambda = \gamma = 0$

Assuming $\boxed{I_1 = 10\mu A}$

→ We know that gain of the above circuit is $(g_{m2})(g_{m6})$

$$10 \times 10^{-6} = \frac{1}{2} \times 230 \times 10^{-6} \times \left(\frac{\omega}{L}\right)_1 (0.2)^2$$

$$\Rightarrow \left(\frac{\omega}{L}\right)_1 = 2.17$$

On solving for ②, we will get $\left(\frac{\omega}{L}\right)_2 = \left(\frac{\omega}{L}\right)_1 = 2.17$

Solving for ③ & ④ -:

$$10 \times 10^{-6} = \frac{1}{2} \times 100 \times 10^{-6} \times \left(\frac{\omega}{L}\right)_{3,4} (0.2)^2$$

$$\Rightarrow \left(\frac{\omega}{L}\right)_{3,4} = 5$$

Solving for ⑤ -:

$$20 \times 10^{-6} = \frac{1}{2} \times 230 \times 10^{-6} \times \left(\frac{\omega}{L}\right)_5 (0.2)^2$$

$$\Rightarrow \left(\frac{\omega}{L}\right)_5 = 4.347$$

On solving for ⑥, we will get $\left(\frac{\omega}{L}\right)_6 = \left(\frac{\omega}{L}\right)_3 = 5$

Solving for ⑦ -:

$$2 \times 10^{-6} = \frac{1}{2} \times 230 \times 10^{-6} \times \left(\frac{\omega}{L}\right)_7 (0.2)^2$$

$$\Rightarrow \left(\frac{\omega}{L}\right)_7 = 0.434$$

→ I took Length of all transistors same and equal to $1\mu\text{m}$.

$$\Rightarrow W_1 = W_2 = 2.17\mu\text{m}$$

$$W_3 = W_4 = W_6 = 5\mu\text{m}$$

$$W_7 = 0.434\mu\text{m}$$

$$W_5 = 4.34\mu\text{m}$$

On simulating the circuit on LTSpice my circuit just fell short of the required gain.

** To improve the gain, I changed the aspect ratio of ⑥ $\Rightarrow \left(\frac{W}{L}\right)_6 = g \Rightarrow W_6 = 9\mu\text{m}$

→ After the following procedure, over all the gain conditions were met!

$$R_{0,1} = 1.13 \times 10^6 \Omega$$

$$R_{0,2} = 1.14 \times 10^6 \Omega$$

$$R_{0,3} = 1.13 \times 10^6 \Omega$$

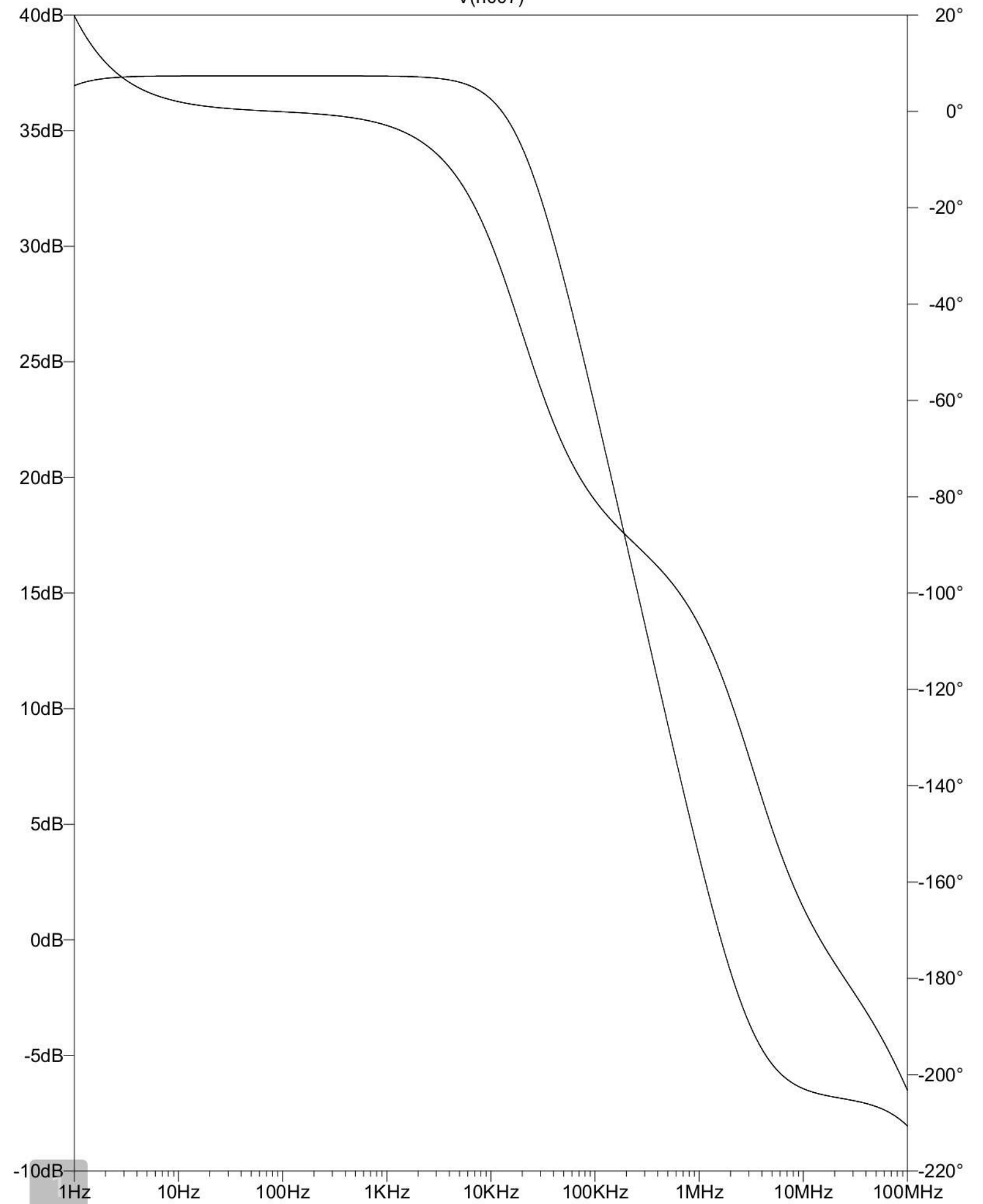
$$R_{0,4} = 1.14 \times 10^6 \Omega$$

$$R_{0,5} = 5.68 \times 10^5 \Omega$$

$$R_{0,6} = 2.61 \times 10^5 \Omega$$

Initial DC Gain Bode Plot

V(n007)



$$V_{DS,2} = 0.83V$$

$$V_{DS,1} = 0.685V$$

$$|V_{DS,3}| = 0.807V$$

$$V_{GS,1} = V_{GS,2} = 0.592V$$

$$V_{GS}, V_{OV,1} = V_{OV,2} = 0.222V$$

$$|V_{OV,3}| = V_0 = 0.603V$$

$$|V_{OV,4}| = 0.28V$$

$$|V_{DS,4}| = 0.662V$$

$$V_{DS,5} = 0.308V$$

$$V_{GS,5} = 0.595V$$

$$|V_{OV,5}| = 0.28V$$

$$V_{OV,5} = 0.308V$$

$$\rightarrow |I_d(M_6)| = 38.2 \mu A$$

$$\bullet |I_d(M_4)| = 8.75 \mu A$$

* $I_d(M_4) \neq I_d(M_6)$ as we changed the aspect ratio of M_6 to increase the gain.

$$\bullet |I_d(M_3)| = 8.84 \mu A$$

$$\bullet I_d(M_1) = 8.84 \mu A$$

$$\bullet I_d(M_2) = 8.76 \mu A$$

$$\bullet I_d(M_5) = 17.6 \mu A$$

$\rightarrow$ The Power dissipation of the circuit came out to be $= 90 \times 10^{-6}$ Watt

$$g_{m1} = 7.96 \times 10^{-5} S$$

$$g_{m2} = 7.89 \times 10^{-5} S$$

$$g_{m3} = 2.93 \times 10^{-5} S$$

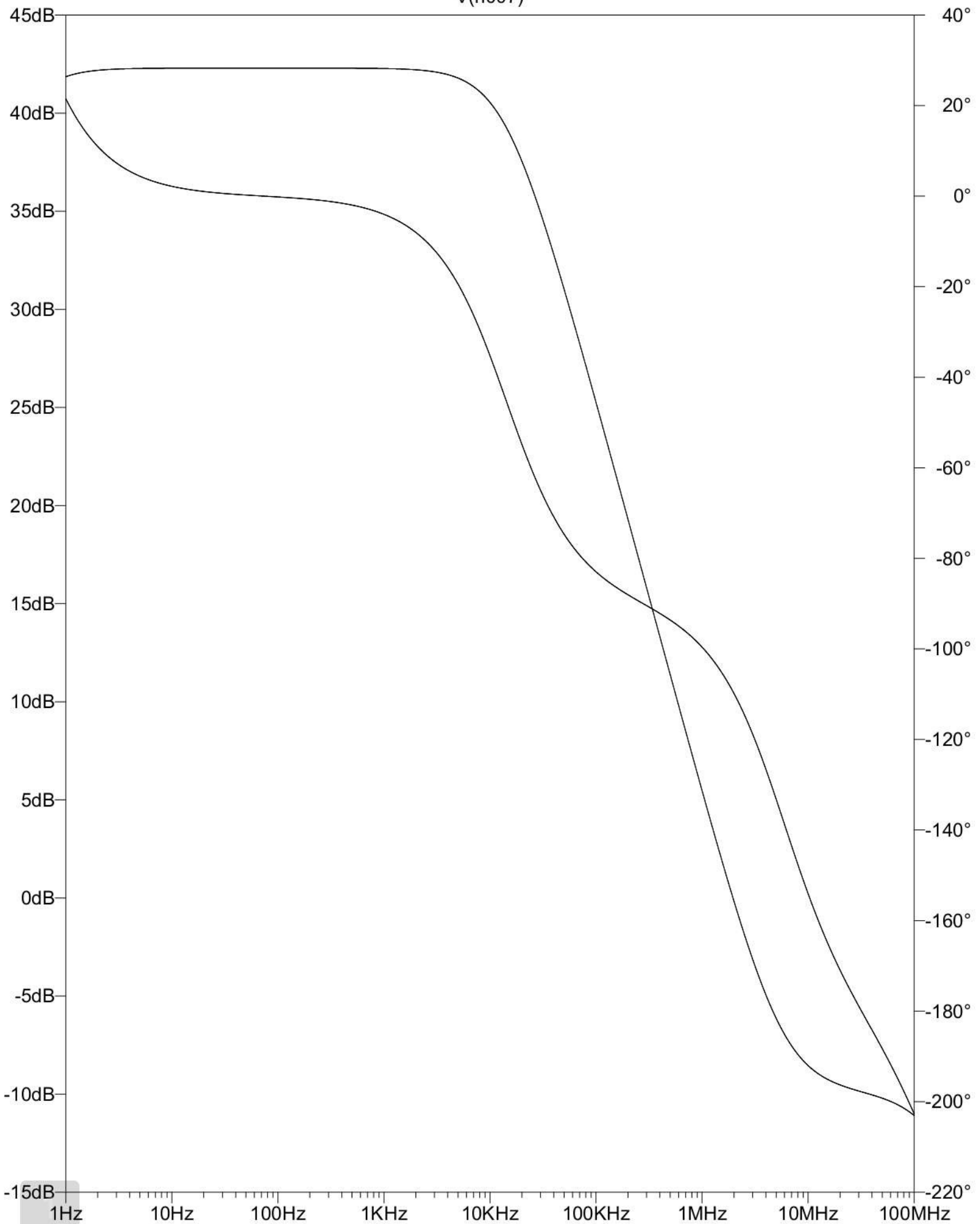
$$g_{m4} = 6.18 \times 10^{-5} S$$

$$g_{m5} = 11.42 \times 10^{-5} S$$

$$g_{m6} = 27.28 \times 10^{-5} S$$

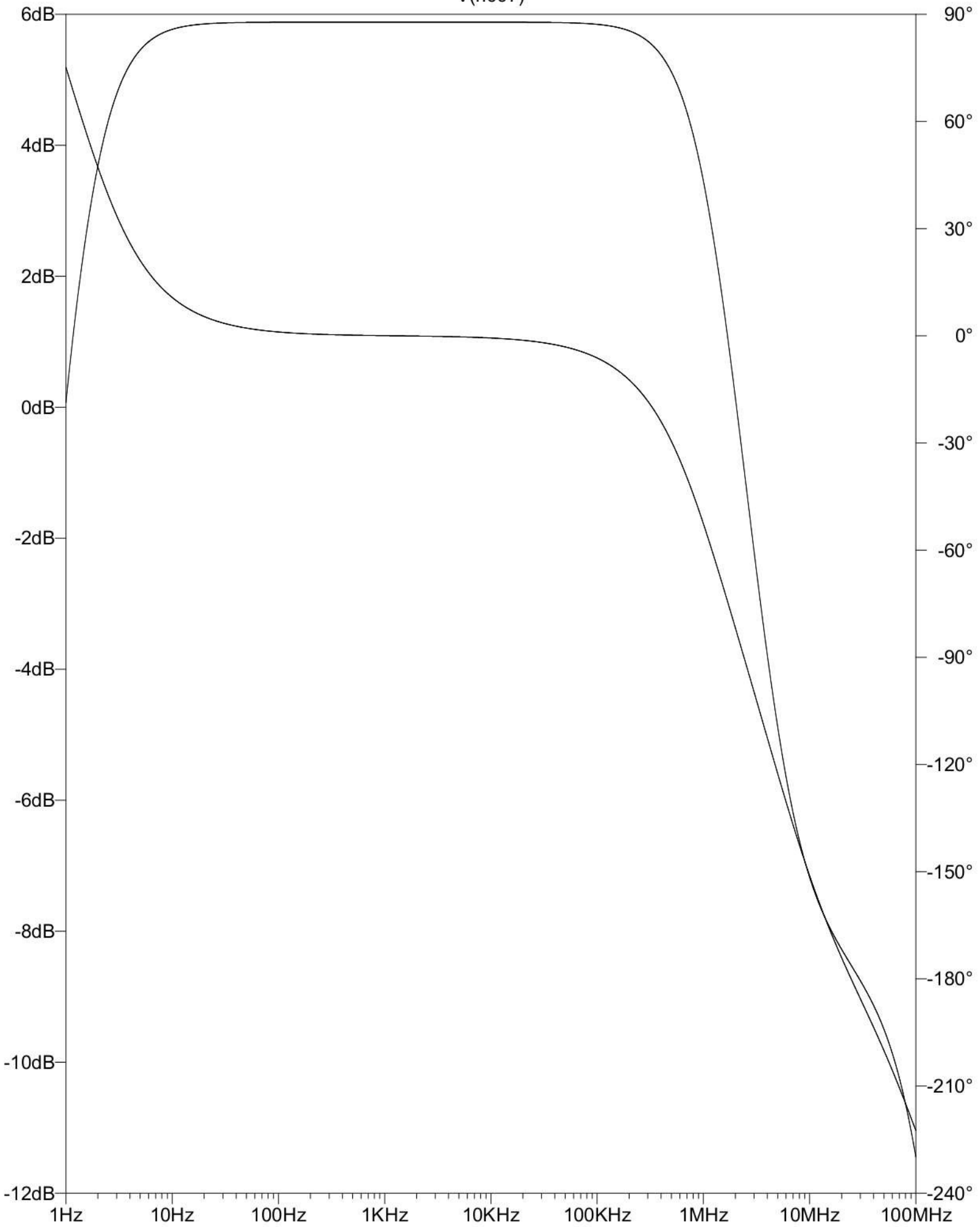
Improved DC Gain Bode Plot

V(n007)



AC Gain Bode Plot

V(n007)



Transient Response

V(n007)

